Text Classification of Movie Dialogue: Comparing Model Performance

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1. Data Preparation

Before training anything, I read through the provided CSVs and found three problems. There were 0 DARTH VADER lines (the Cornell corpus calls him `VADER`), lines from same-named characters in other films were mixed in (e.g., the EMPEROR from *Dune*), and the "1000lines" file had only 552 rows. So I re-extracted only the three Star Wars films by movie ID and removed duplicates. The corpus has only about 490 such lines, so 1,000 was not possible.

Dataset	LUKE	LEIA	VADER	EMPEROR	Total	Bad share
Small (Part 1)	94	–	25	6	125	24.8%
Large (Part 2)	229	175	62	14	480	15.8%
Provided Large (comparison)	340	174	0	23	537	4.3%

Every setup used the same repeated stratified 5-fold CV (25 folds) inside a leak-free Pipeline.

2. Results

Means over 25 folds; Macro-F1 is mean ± SD.

Data	Vectorizer	Model	Imbalance	Acc	Bad P	Bad R	Bad F1	Macro-F1
Small	TF-IDF	LogReg	balanced	0.813	0.660	0.567	0.593	0.735 ± 0.107
Small	SBERT	LogReg	balanced	0.858	0.735	0.708	0.710	0.807 ± 0.088
Large	TF-IDF	LinearSVC	balanced	0.890	0.716	0.537	0.605	0.771 ± 0.035
Large	TF-IDF	LogReg	balanced	0.876	0.622	0.587	0.599	0.763 ± 0.040
Large	SBERT	LogReg	balanced	0.847	0.522	0.722	0.599	0.752 ± 0.046
Large	Count	MultinomialNB	none	0.896	0.826	0.437	0.560	0.751 ± 0.070
Large	Count	GradBoost	balanced	0.856	0.588	0.507	0.528	0.721 ± 0.064
Large	Count	RandomForest	none	0.870	0.888	0.198	0.315	0.622 ± 0.074

3. Answers to the Research Questions

Q1. Data scale: With more data, Macro-F1 went up and its SD went down in 19 of 20 setups (final model SD 0.107 → 0.040). Bad Recall, though, barely moved (0.567 → 0.587).

Q2. Accuracy vs. Bad Recall: The most accurate model on Large, Count + NB (0.896), caught only 0.437 of Bad lines. Unweighted TF-IDF LR scored 0.842 accuracy while calling almost everything Good (Bad Recall 0.003). Accuracy alone misleads on imbalanced data.

Q3. Count vs. TF-IDF: TF-IDF helped the balanced linear models (+0.03–0.05 Macro-F1), while Count was better for NB and RF. Lines are short (median 6 words), so counts are mostly 0/1; I think L2 normalization mattered more than IDF.

Q4. Final model: I chose TF-IDF + Logistic Regression (balanced). It ranked 1st on Small (0.735) and 2nd on Large (0.763), with both Bad Precision and Recall above 0.5.

Q5. Pre-trained embeddings: I ran SBERT (all-MiniLM-L6-v2) + LR (balanced) in Colab on the same folds. It beat TF-IDF on Small (Macro-F1 0.807 vs. 0.735). On Large, higher Bad Recall (0.722 vs. 0.587) but lower Precision (0.522) left Macro-F1 similar (0.752). At ~12 ms per line, it is worth it for small data.

4. Misclassified Examples

The final model (Large) missed 33 of 76 Bad lines and flagged 24 Good lines, relying on topic words and the villains' commanding style ("the").

Actual → Predicted	Speaker: Line	Cause (contributing terms)
Bad → Good	VADER: What is thy bidding, My Master?	what (+0.28) offsets my master (−0.41)
Good → Bad	LUKE: Lord Vader … Is the dark side stronger?	topic word dark side (−0.23)
Good → Bad	LEIA: Yes.	yes (−0.74), from Vader's lines

5. Limitations and What I Learned

- **Limitations:** With only 76 Bad lines, the top models can't be separated statistically, and lines like "Yes." can't be classified by character labels. Next time I would add nested CV and more villain characters.
- **What I learned:** Working with Korean course materials and English data as an international student, I learned to check the data myself before trusting any model.